

Legislative Coordination and Institutional Change

A Bipartite Network Analysis of Coalition Dynamics in the Brazilian Chamber of Deputies

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19th Political Networks Conference (POLNET 2026) · University of Manchester

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Slides & paper

| The puzzle

- ◆ Polarization should make legislatures **ungovernable**.
- ◆ Brazil 2014–16: impeachment · corruption scandals · recession · polarization.
- ◆ Yet the Chamber **kept forming majorities**, across impeachment and a change of power.
- ◆ The literature's crisis typology: **collapse · fragmentation · realignment · institutionalization**. Brazil fits **none**.
- ◆ **How? And at what cost?**

Coalitional presidentialism, and a cluster of shocks

- ◆ Strong president, **no majority** → govern by **coalition** (Abranches 1988).
- ◆ Levers: **budget · posts · agenda**; the **Centrão** sells support as the president's *agent*.
- ◆ Recent work reads a **weakening presidency and a rising Congress**, much of it via **budget powers** once a pillar of the president's toolkit.
- ◆ 2014–16, all at once: impeachment · corruption scandals · recession · **and the president's budget lever removed**.
- ◆ **Shocks co-occur in a tight 2014–16 window** → no single lever separates cleanly from the rest (hence the eliminative design).

| The structural question, and some rival explanations

- ◆ The open question is structural: **How did coordination reorganize after the 2014–2016 shocks? What happened to the pragmatic center, and at what cost?**
- ◆ Then rule out the rival explanations, one at a time:
- ◆ Is it **turnover** (new members) or a one-off **crystallization**?
- ◆ Is it the **Cunha effect** (the agenda-setter) or the **fiscal channel**: does re-sorting concentrate on economic-fiscal rather than rights-and-moral bills?

The tests ask whether a durable pivot remains once turnover, agenda, and fiscal explanations are removed.

Why networks, not ideal points

- ◆ **deputies × bills** → agreement graph → **communities**.
- ◆ Standard **one-dimensional ideal-point** models compress change onto one axis.
- ◆ Blocs **recompose** while ideology looks stable; an axis can't see it.
- ◆ No axial assumption → the shift is **bloc recomposition**.

Related work, and how this differs

- ◆ **Prior networks of the Chamber exist** (Brito 2020; Toledo Jr. 2025). **Four differences:**
- ◆ **1. Bipartite primitive:** starts from the deputy×bill incidence and explicitly tests projection dependence.
- ◆ **2. Null-corrected structure:** modularity against a **within-vote permutation null (Q^*)**, not a raw score.
- ◆ **3. Within-legislator design:** separates a change in behavior from a change in composition.
- ◆ **4. Theory-driven object:** the object is **which coordinating community a bloc occupies relative to the executive**, not a left–right axis.

| The methodological approach

- ◆ **Measure:** roll-calls → **deputy × bill** graph → **communities per legislature**.
- ◆ **Classify bills:** fiscal vs moral, **rule-based** + local **LLM cross-check** (Qwen 2.5 32B), human-validated (n=294, $\alpha=0.84$, macro-F1 0.88, 89.5%).
- ◆ **Compare:** across the 2014–16 break; a **within-deputy panel** separates behavior from turnover.
- ◆ **Falsify, not estimate:** near-collinear shocks → **eliminate rivals one at a time**.
- ◆ **Stress-test:** Q^* null, 12-cell threshold grid, alternative projections and algorithms.

Votes as a bipartite incidence matrix

$$X_{ij} = \begin{cases} +1 & \text{deputy } i \text{ voted } \mathbf{Yes} \\ -1 & \text{voted } \mathbf{No} \\ 0 & \text{abstention, obstruction, or absent} \end{cases}$$

Deputy × bill, before projecting to the agreement graph.

The coordination metric: agreement score

$$W_{ij} = \frac{|\{v : S_{iv} = S_{jv} \neq 0\}|}{|\{v : S_{iv} \neq 0 \text{ and } S_{jv} \neq 0\}|}, \quad S = \text{sign}(X)$$

where W_{ij} = agreement rate between deputies i and j • $S = \text{sign}(X)$ = signed vote matrix (+1 Yes, −1 No, 0 otherwise) • v = a recorded vote

numerator = votes where i, j agree • denominator = votes both attended • edge set to 0 if fewer than 30 shared votes

Krehbiel (1993) co-voting agreement, normalised by joint attendance.

The ERGM specifications I tried

$$\Pr(Y = y) = \kappa^{-1} \exp\left(\sum_k \theta_k z_k(y)\right)$$

- 1 **bipartite pre-test** z : edges + gwb1deg + gwb2deg DID NOT MIX
- 2 **cross-sectional** z : edges + nodematch(party, Centrao, gov/opp, UF) RUNS
- 3 **temporal (btergm)** z_t : edges + gwesp + homophilies + memory RUNS

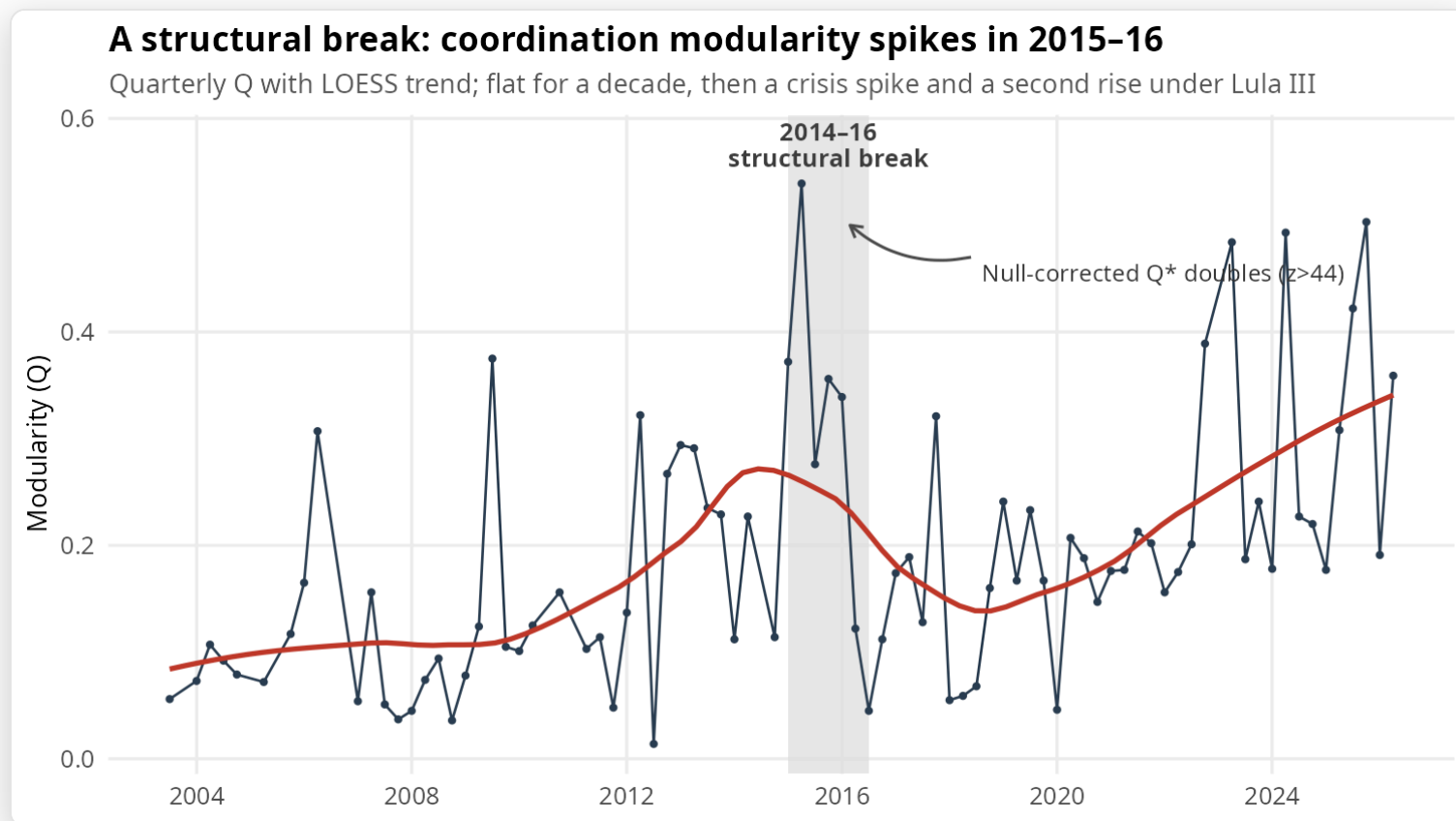
statistics edges = $\sum_{i < j} y_{ij}$ • nodematch_x = $\sum_{i < j} y_{ij} \mathbf{1}[x_i = x_j]$ • gwesp = geom.-weighted shared partners (closure) • memory = tie persistence across t

Same estimand throughout, the marginal effect of EC 86; the binding constraint is identification, not mechanics.

What the ERGM exercises establish

- ◆ Estimand: the **causal effect of mandatory amendments** (EC 86).
- ◆ **Static bipartite ERGM** (deputy×bill incidence): **did not mix** on the dense 54L network.
- ◆ **Temporal agreement-network model** (btergm): **runs**, recovering **closure** and **tie persistence**.
- ◆ **Identification remains:** neither exercise separates EC 86 from the post-2014 shock cluster (VIF 11.6).
- ◆ **Implication:** estimation can *model* coordination, but cannot *isolate* the reform's marginal effect.

A structural break: modularity spikes in 2015–16



Quarterly modularity rises from $\approx .15$ to $.51$ in 2016.

Null-corrected modularity, Q^*

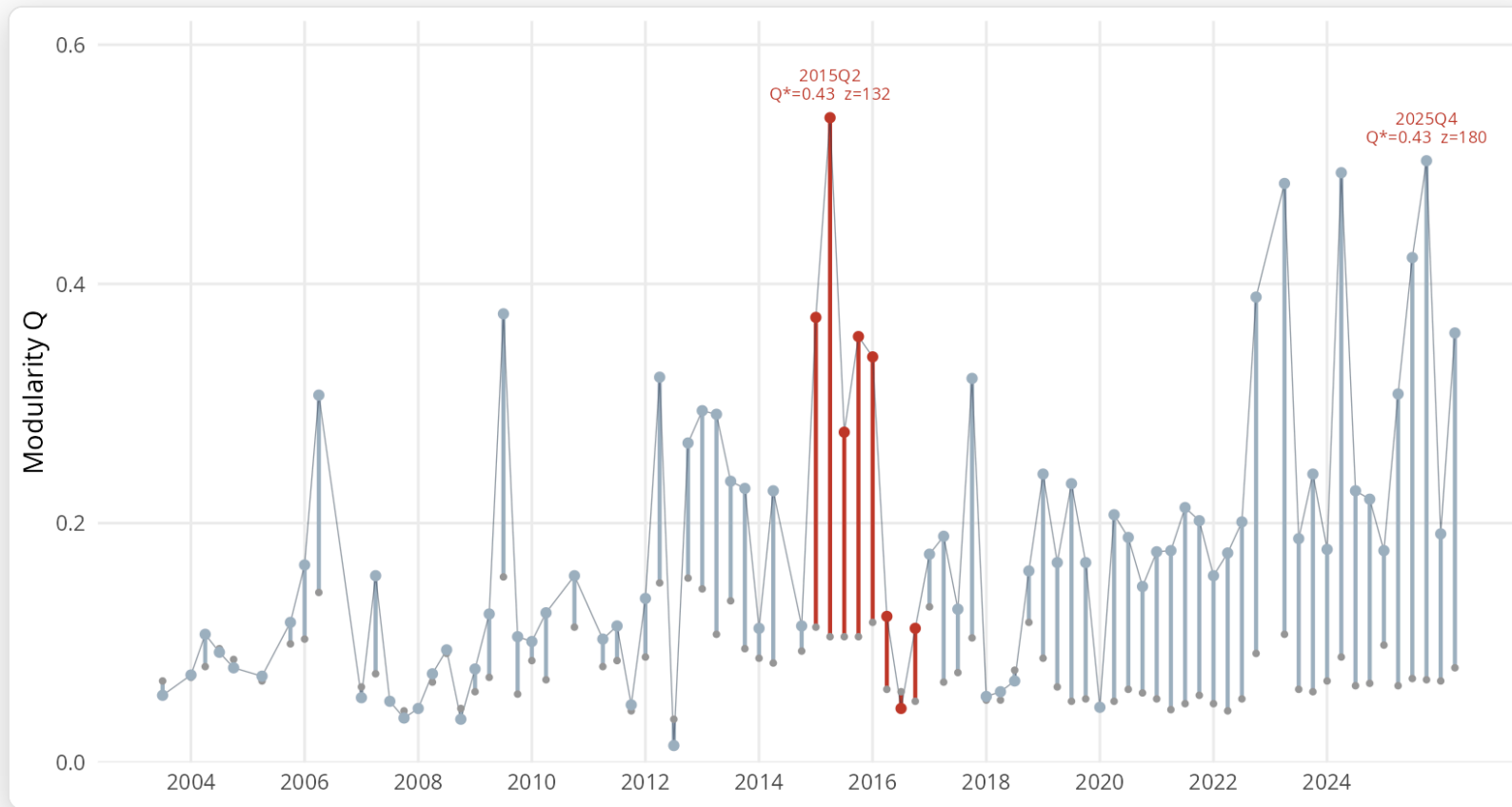
$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

$$Q^* = Q_{\text{obs}} - \langle Q_{\text{null}} \rangle, \quad z = \frac{Q_{\text{obs}} - \mu_{\text{null}}}{\sigma_{\text{null}}}$$

where A_{ij} = agreement-graph weight • k_i = weighted degree of i • m = total edge weight • $\delta(c_i, c_j) = 1$ if i, j share a community, else 0 • null = within-vote permutations

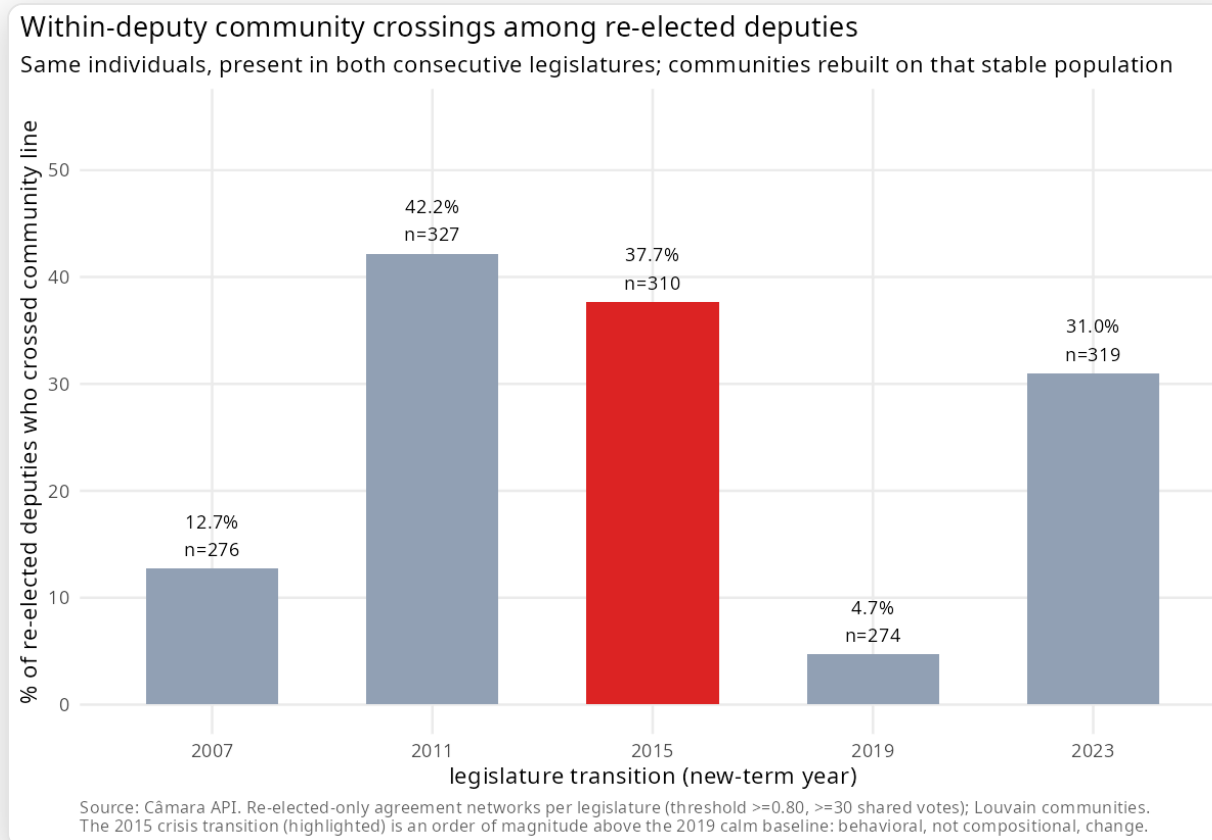
Newman–Girvan modularity, benchmarked against a permutation null: Q^* is what exceeds chance.

Null-corrected modularity (Q^*) over time



Observed Q exceeds the within-vote null, peaking in 2015–16 and rising again under Lula III.

Same people, not turnover

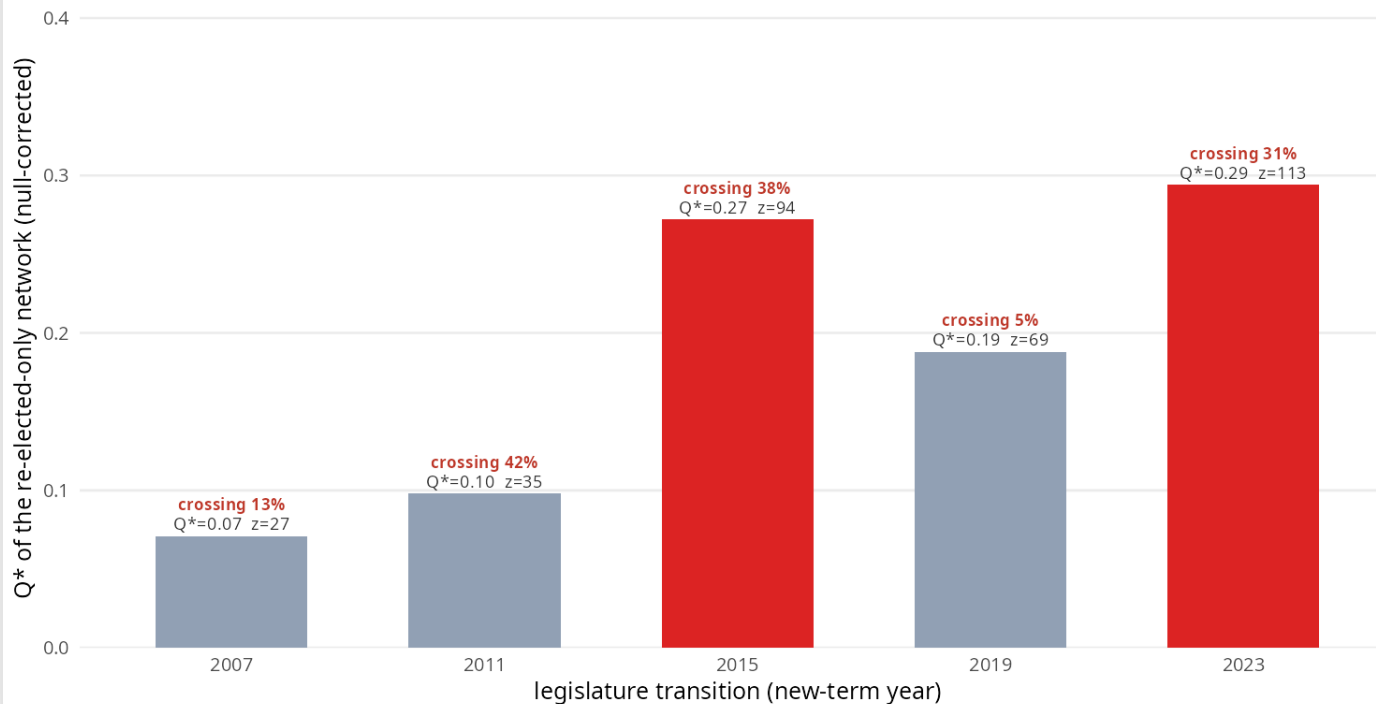


Reelected-deputy crossing: **37.7%** at the break, **4.7%** afterward.

Movement is not crystallization

Same people: movement vs crystallization

Null-corrected modularity among deputies present in both legislatures. Crossing is movement, Q^* is structure. The two re-sorts (2015 crisis, 2023 Lula III alternation) move the same people AND sharpen the structure; 2011 churns (high crossing, low Q^*) and 2019 holds (consolidated, little movement).

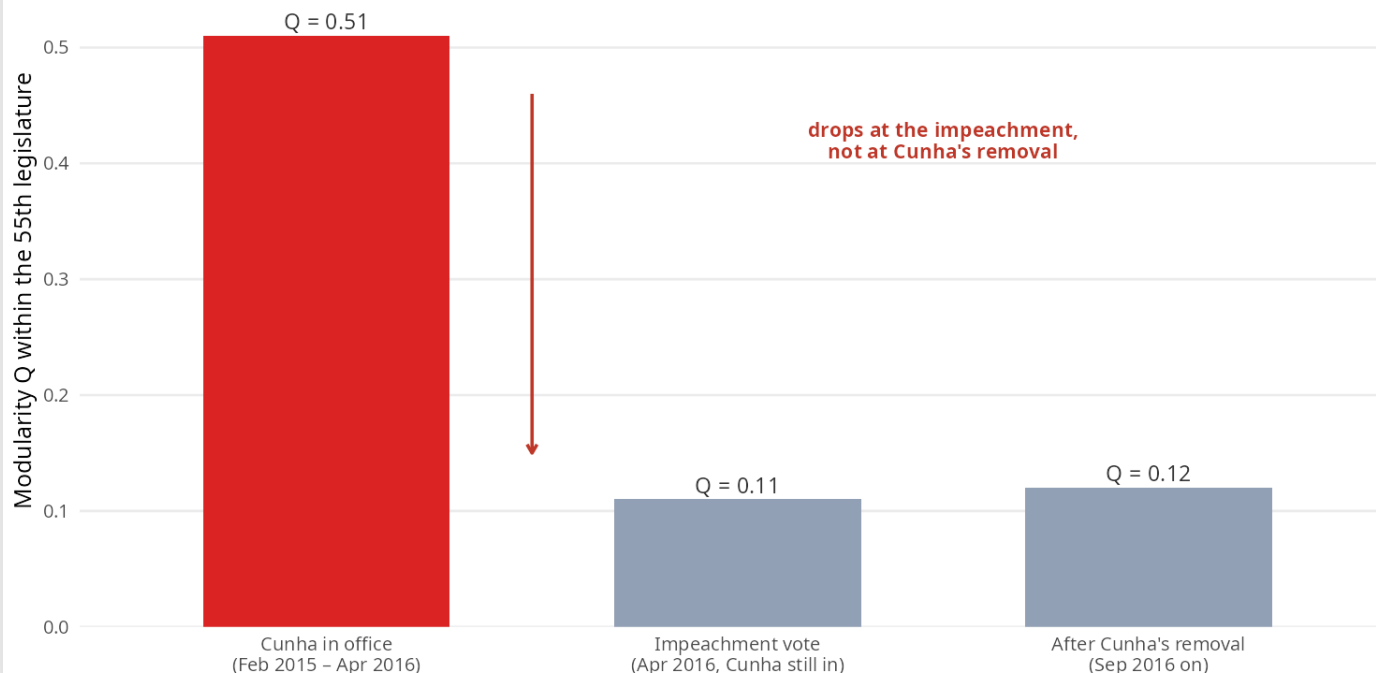


2011 moves more deputies but stays diffuse; 2015 and 2023 sharpen the structure.

Not Cunha's agenda: dated by the impeachment

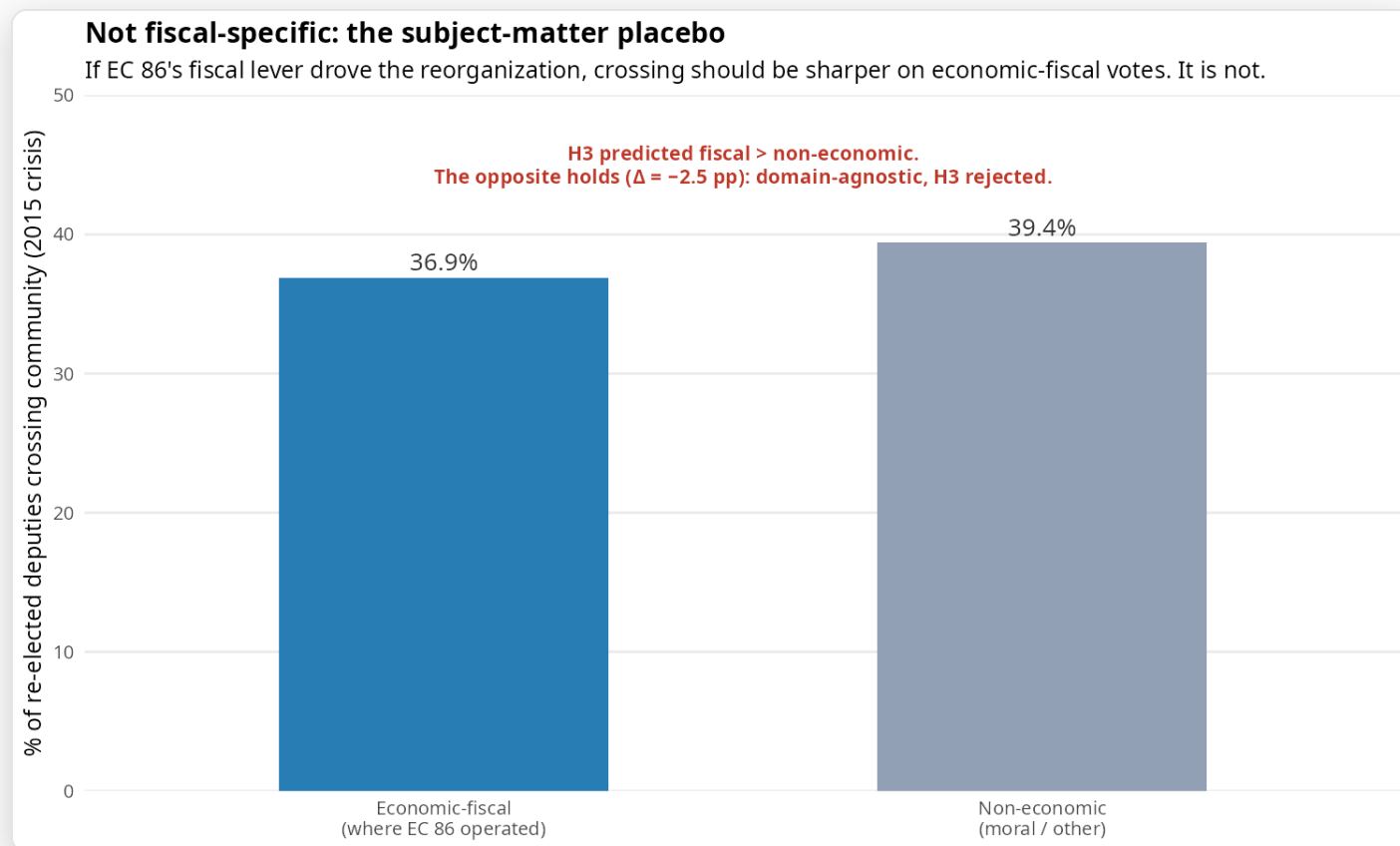
Not Cunha's agenda: the recovery is dated by the impeachment

If Cunha's agenda control were the mechanism, Q would stay high until he left (Sep 2016). It falls at the April-2016 impeachment, while Cunha is still in office: the polarization episode, not the individual.



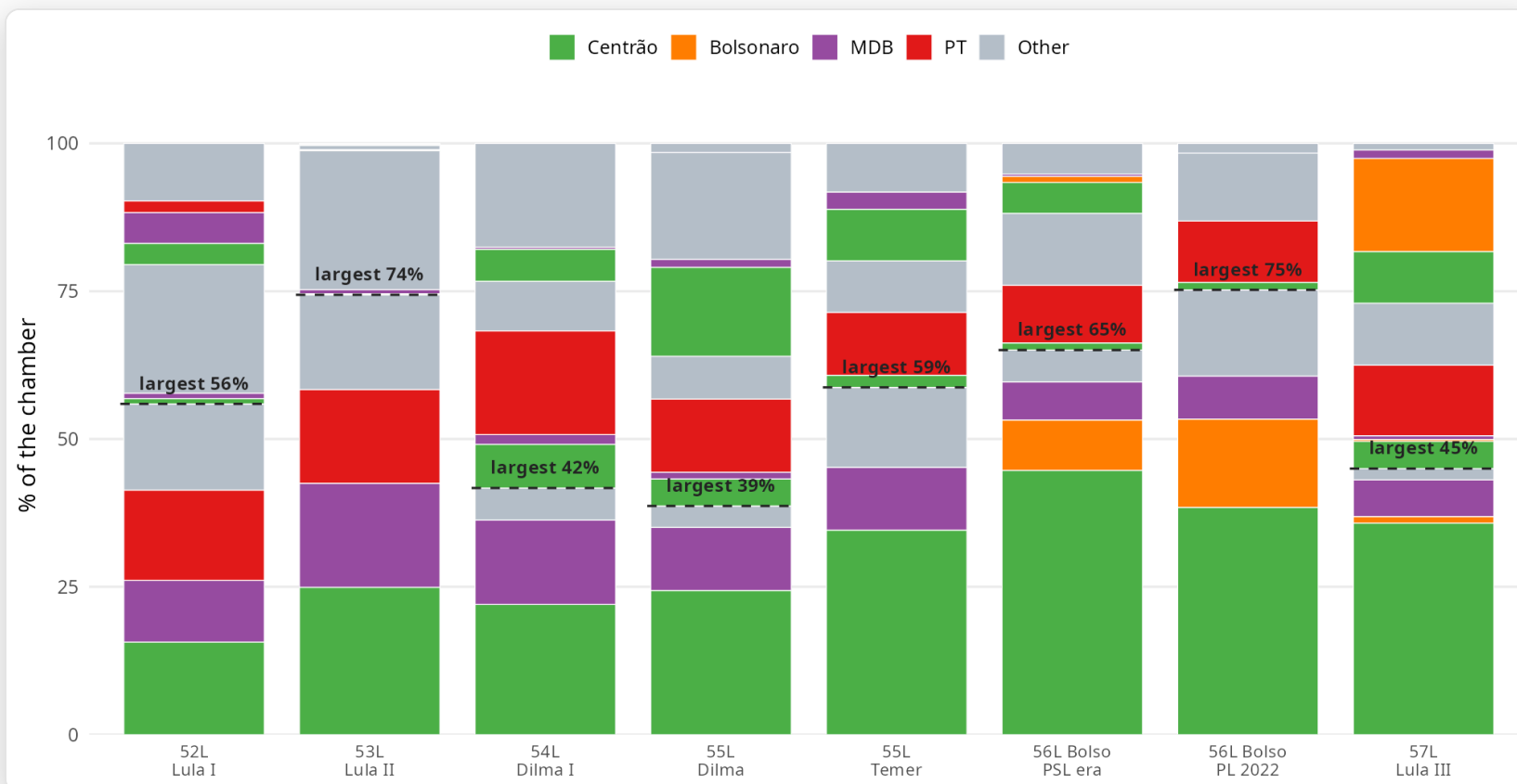
Modularity falls at the impeachment, not at Cunha's removal.

Not fiscal-specific: the subject-matter placebo



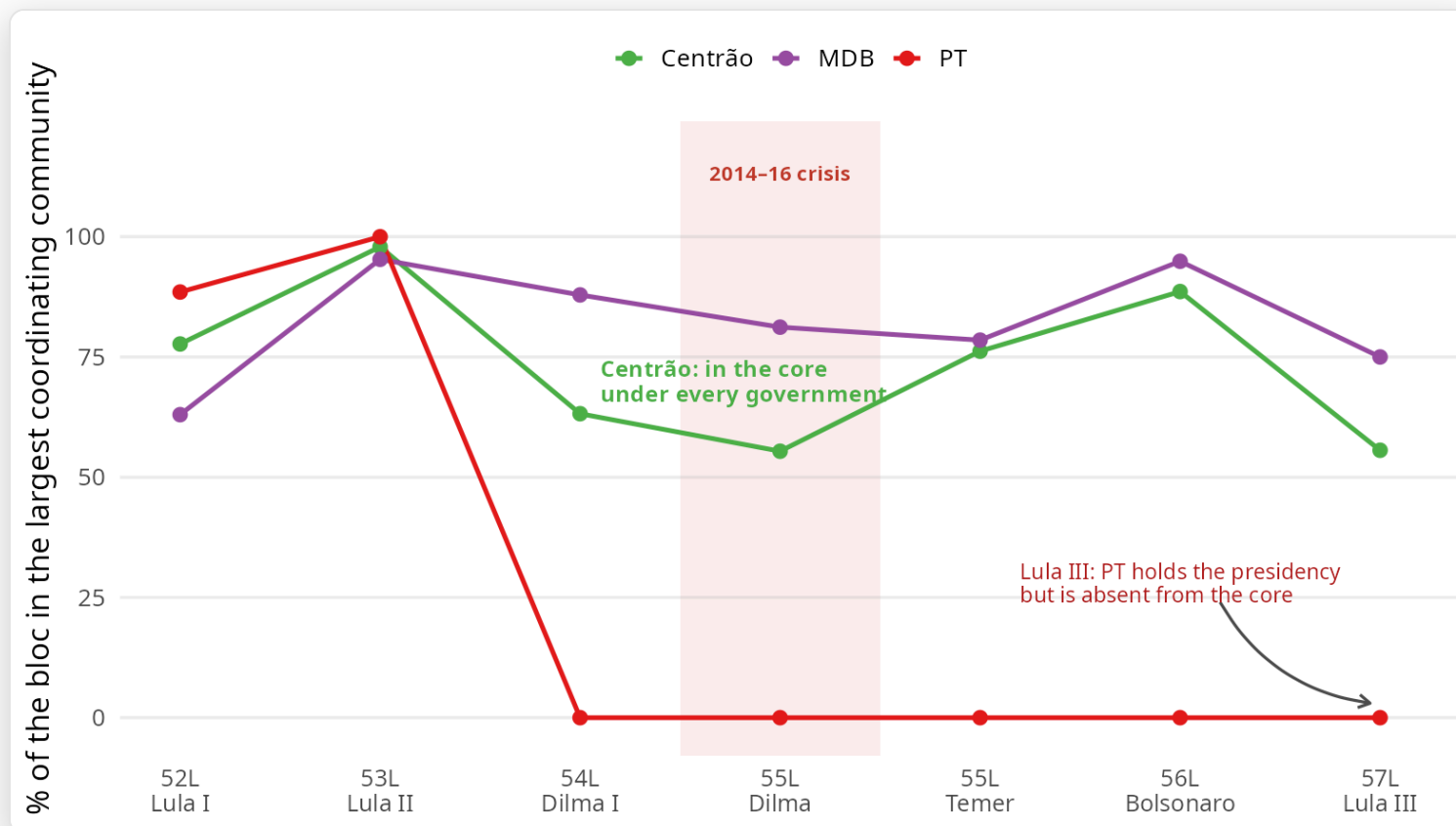
Fiscal: 36.9%; non-economic: 39.4%; $\Delta = -2.5$ pp.

The coordinating communities, sized



From 2014, the Centrão anchors the largest community while the PT moves to a smaller one.

The pivot: in the largest community under every government

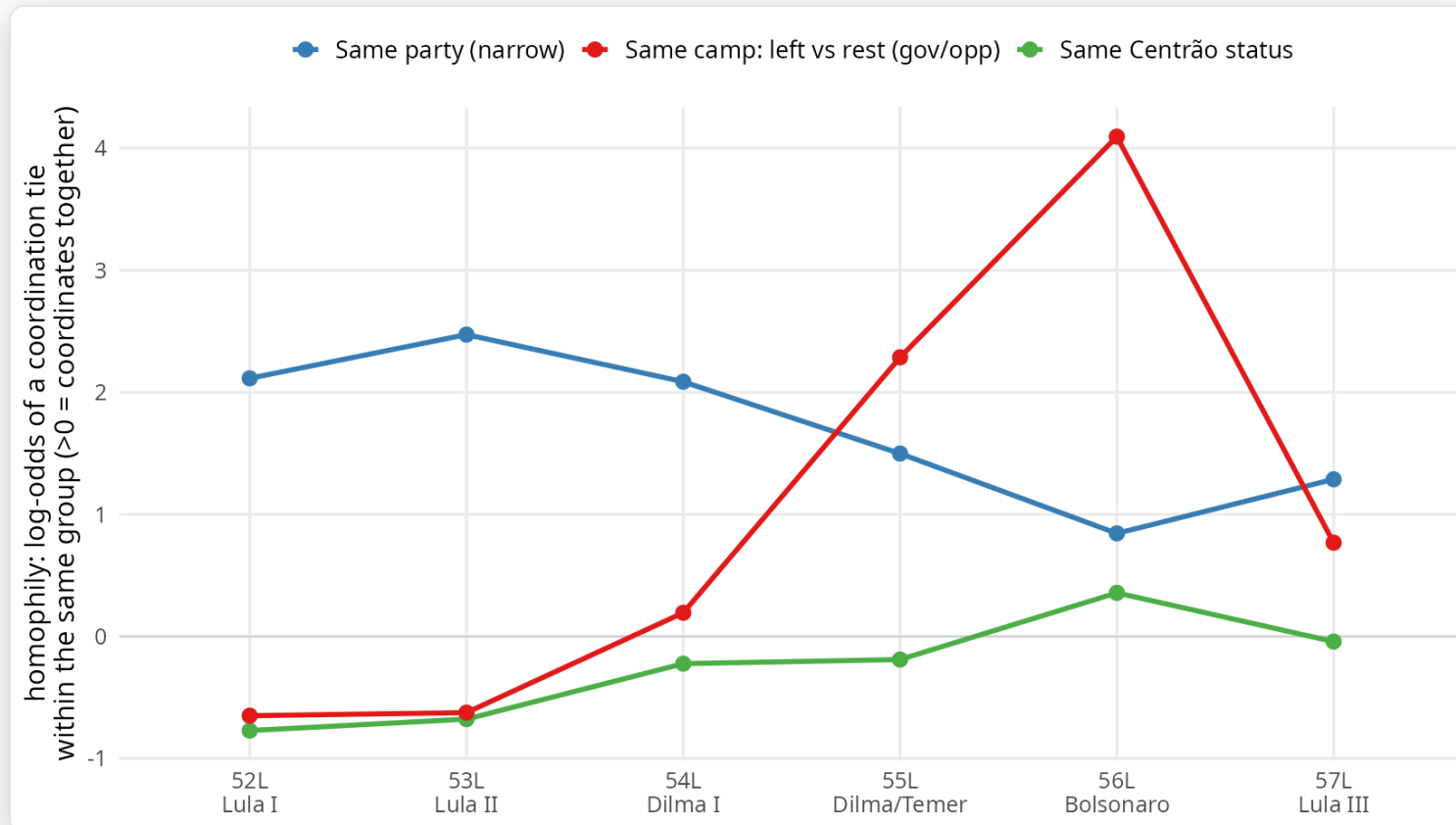


The Centrão remains in the largest community across governments; the PT does not.

Can we trust these communities?

- ◆ **Noise?** Q^* sits far above a within-vote permutation null ($z > 44$).
- ◆ **Projection?** A projection-free bipartite SBM shows compatible alignment at the 56L (ARI 0.23), weakening the projection-artifact interpretation.
- ◆ **Labels?** Vote-only communities increasingly align with an independently coded Centrão (ARI $0 \rightarrow 0.39$).
- ◆ **Model dependence?** A descriptive ERGM detects the same shift, from party homophily to government–opposition organization.
- ◆ **Just the PT in opposition?** No: the government side organizes as strongly as the opposition, and the shift survives dropping the PT.

What organizes coordination: party gives way to position



Party homophily declines as government–opposition organization rises after 2014.

| The thesis, and what it means

- ◆ The pragmatic center is now the recurring pivot governments build majorities through, coordinating by position relative to the executive, not a fixed ideological axis.
- ◆ The Centrão is old (1987); what changed is its position, *agent* → *pivot*.
- ◆ Why (a reading, not a tested cause): the budget shifted to Congress (mandatory amendments, then the secret budget and agenda control), so the tool that once disciplined the pivot now lets it set the terms (*governo congressual*).
- ◆ What it means: governability sold, not represented; the cost is accountability, not gridlock.

Contribution, and what the case teaches

- ◆ **Measurement:** recover the Centrão's permanence from the votes, its position and persistence, where the literature had narrative.
- ◆ **Design:** with near-collinear shocks, eliminate the rival explanations rather than estimate one effect.
- ◆ **A portable template:** turn a narrative prediction, that institutional crisis reshapes the legislature, into a measurable, falsifiable object, and narrow the viable mechanisms.
- ◆ **A way of looking, not a new claim:** the permanent-pivot type is old (Israel, post-war Italy).

Next steps

- ◆ **Separate influence from selection:** a contagion-vs-similarity REM (Leifeld et al. 2026) or a SAOM; my bipartite data has the structure.
- ◆ **A formal inferential backbone:** a degree-corrected SBM, and a validated backbone (Neal's SDSM) beyond the edge filters.
- ◆ **Quantify evidence of absence:** Bayes factors on the null placebos.
- ◆ **Travel the type:** the permanent governing pivot elsewhere (Israel, post-war Italy).

I'd value your pointers and collaborations on any of these.

Thank you

Questions welcome.

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